

Department of Electronics and Telecommunication Engineering
(NBA ACCREDITED)
Digital System Design Laboratory
Academic Year 2025-2026, SEM-5, Odd Semester

Course Code	EC24P
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Class	T.E – EXTC/EXCS SEM-V
Division	B
Date of Performance	8 th August 2025
Date of Submission	22 nd August 2025

EXPERIMENT NO.4

Estimated time to complete this experiment: 02 hours.

Objective: To simulate and Implement 4:1 Mux & 1:4 D-Mux using all methods of modeling.

CO to be achieved: CO1, CO2

Expected Outcome of Experiment:

- Students will learn to design, simulate and synthesis the arithmetic circuit using VIVADO & VERILOG.
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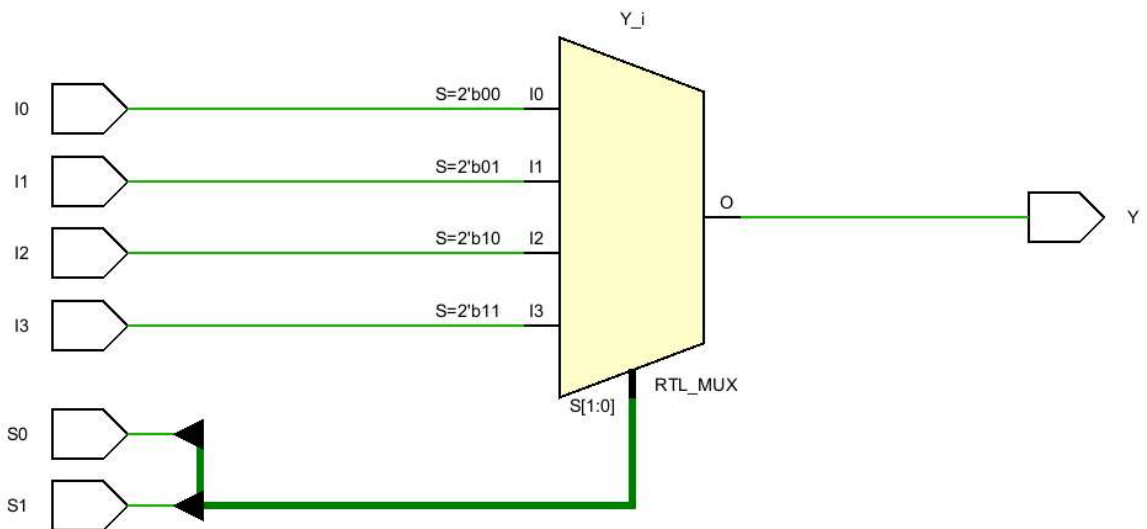
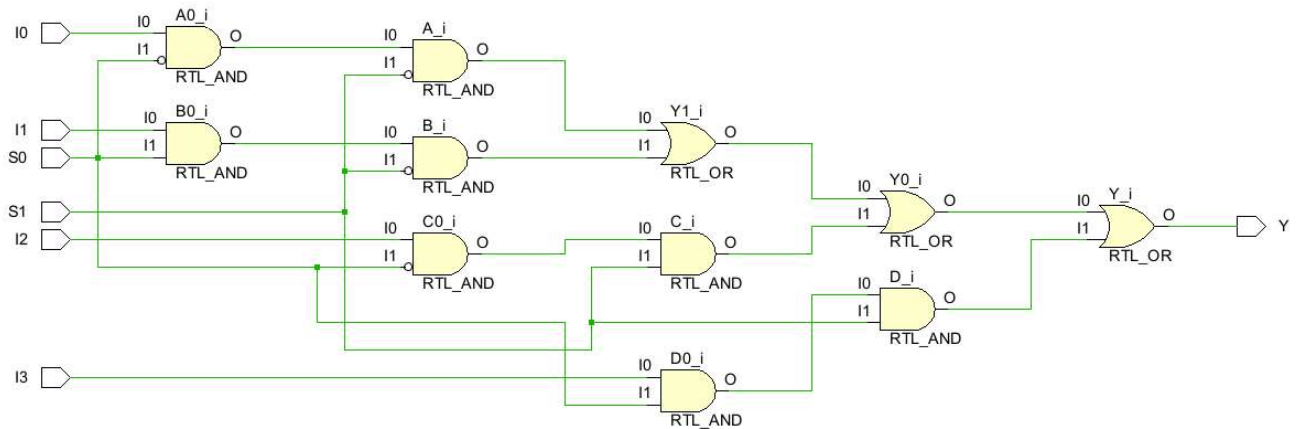
Pre Lab/ Prior Concept: Designing of Mux & D-Mux circuits.

Apparatus/TOOL:

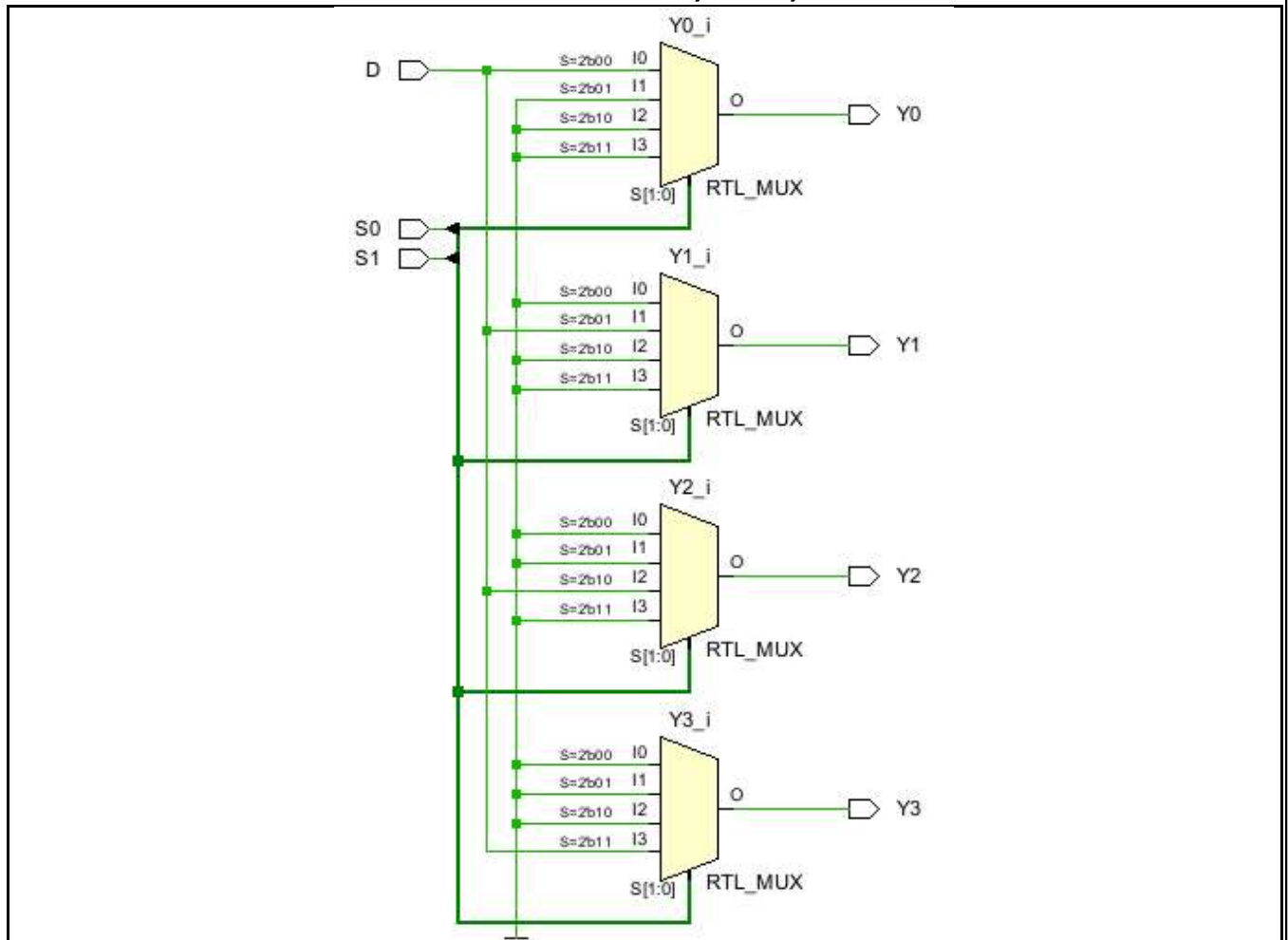
- PC
 - VIVADO TOOL
 - SPARTABN-7 FPGA Board.
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CIRCUIT DIAGRAM.

MUX:



DEMUX:



VERILOG - CODE.

1:4 MULTIPLEXER

1. Gate-Level Modelling

Code

```
module MUX_4to1_Gate_Level(output Y, input I0, I1, I2, I3, input S0, S1);
    wire S0_n, S1_n, A, B, C, D;

    // Generate inverted select signals
    not (S0_n, S0);
    not (S1_n, S1);

    // AND gates for each input line
    and (A, I0, S0_n, S1_n);
    and (B, I1, S0, S1_n);
    and (C, I2, S0_n, S1);
    and (D, I3, S0, S1);
```

```
// OR gate to combine the outputs  
or (Y, A, B, C, D);  
endmodule
```

Test Bench

```
module MUX_4to1_Gate_Level_tb;  
    reg I0, I1, I2, I3, S0, S1;  
    wire Y;  
  
    // Instantiate the MUX module  
    MUX_4to1_Gate_Level uut (.Y(Y), .I0(I0), .I1(I1), .I2(I2), .I3(I3), .S0(S0), .S1(S1));  
  
    initial begin  
        $monitor("S1 = %b, S0 = %b, I3 = %b, I2 = %b, I1 = %b, I0 = %b, Y = %b", S1, S0, I3, I2, I1, I0, Y);  
  
        // Test cases  
        I0 = 0; I1 = 0; I2 = 0; I3 = 0;  
        S1 = 0; S0 = 0; #10;  
  
        I0 = 1; S1 = 0; S0 = 0; #10;  
        I1 = 1; S1 = 0; S0 = 1; #10;  
        I2 = 1; S1 = 1; S0 = 0; #10;  
        I3 = 1; S1 = 1; S0 = 1; #10;  
  
        $stop;  
    end  
endmodule
```

2. Dataflow Modeling

Code

```
module MUX_4to1_Dataflow(output Y, input I0, I1, I2, I3, input S0, S1);  
    assign Y = (~S1 & ~S0 & I0) |  
        (~S1 & S0 & I1) |  
        (S1 & ~S0 & I2) |  
        (S1 & S0 & I3);  
endmodule
```

3. Behavioral Modeling

Code

```
module MUX_4to1_Behavioral(output reg Y, input I0, I1, I2, I3, input S0, S1);  
    always @(*) begin  
        case ({S1, S0})  
            2'b00: Y = I0;  
            2'b01: Y = I1;  
            2'b10: Y = I2;  
            2'b11: Y = I3;  
        endcase  
    end
```

```
2'b11: Y = I3;  
default: Y = 0;  
endcase  
end  
endmodule
```

1:4 DEMULTIPLEXER

1. Gate-Level Modeling

Code

```
module DEMUX_1to4_Gate_Level(output Y0, Y1, Y2, Y3, input D, S0, S1);  
    wire S0_n, S1_n;  
  
    // Generate inverted select signals  
    not (S0_n, S0);  
    not (S1_n, S1);  
  
    // AND gates for each output line  
    and (Y0, D, S0_n, S1_n); // Y0 active when S1=0, S0=0  
    and (Y1, D, S0, S1_n); // Y1 active when S1=0, S0=1  
    and (Y2, D, S0_n, S1); // Y2 active when S1=1, S0=0  
    and (Y3, D, S0, S1); // Y3 active when S1=1, S0=1  
endmodule
```

Test Bench

```
module DEMUX_1to4_Gate_Level_tb;  
    reg D, S0, S1;  
    wire Y0, Y1, Y2, Y3;  
  
    // Instantiate the DEMUX module  
    DEMUX_1to4_Gate_Level uut (.Y0(Y0), .Y1(Y1), .Y2(Y2), .Y3(Y3), .D(D), .S0(S0), .S1(S1));  
  
    initial begin  
        $monitor("D = %b, S1 = %b, S0 = %b, Y0 = %b, Y1 = %b, Y2 = %b, Y3 = %b", D, S1, S0, Y0, Y1, Y2, Y3);  
  
        // Test cases  
        D = 0; S1 = 0; S0 = 0; #10;  
        D = 1; S1 = 0; S0 = 0; #10;  
        D = 1; S1 = 0; S0 = 1; #10;  
        D = 1; S1 = 1; S0 = 0; #10;  
        D = 1; S1 = 1; S0 = 1; #10;  
        $stop;  
    end  
endmodule
```

2. Dataflow Modeling

Code

```
module DEMUX_1to4_Dataflow(output Y0, Y1, Y2, Y3, input D, S0, S1);
  assign Y0 = D & ~S1 & ~S0; // Y0 active when S1=0, S0=0
  assign Y1 = D & ~S1 & S0; // Y1 active when S1=0, S0=1
  assign Y2 = D & S1 & ~S0; // Y2 active when S1=1, S0=0
  assign Y3 = D & S1 & S0; // Y3 active when S1=1, S0=1
endmodule
```

3. Behavioral Modeling

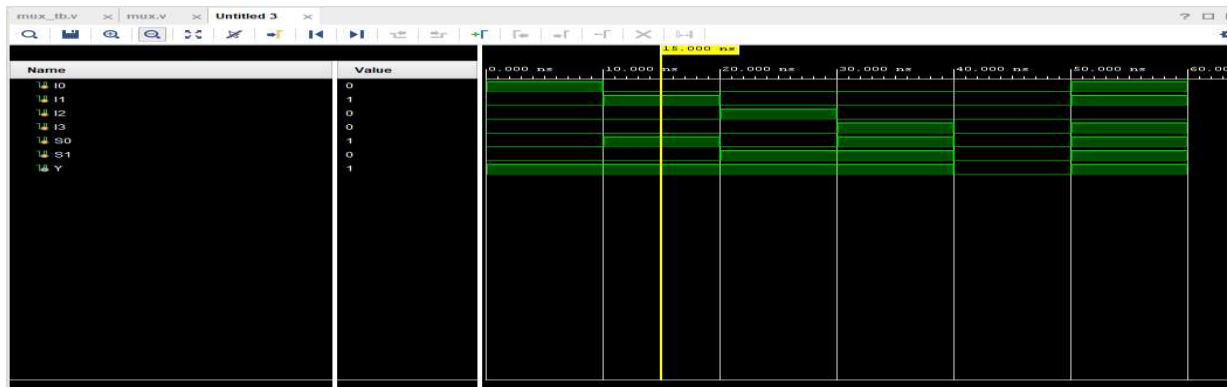
Code

```
module DEMUX_1to4_Behavioral(output reg Y0, Y1, Y2, Y3, input D, S0, S1);
  always @(*) begin
    // Default values
    Y0 = 0; Y1 = 0; Y2 = 0; Y3 = 0;

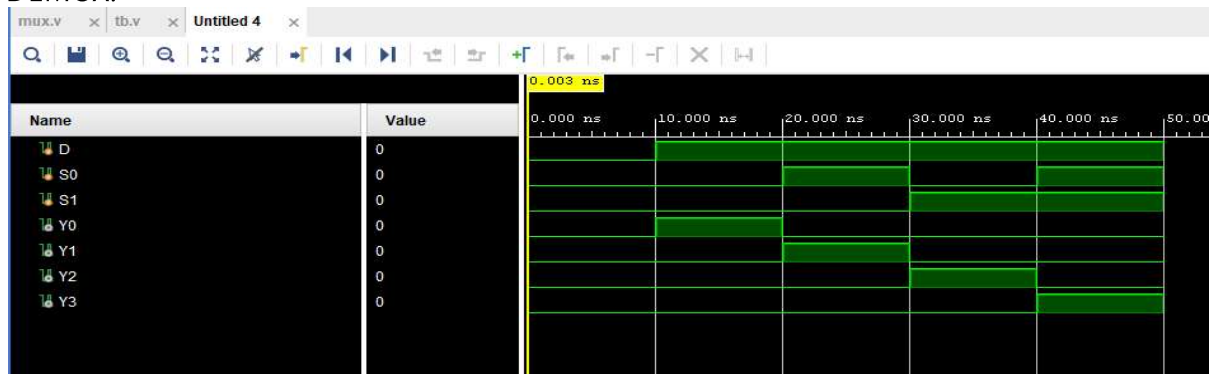
    case ({S1, S0})
      2'b00: Y0 = D; // Y0 active when S1=0, S0=0
      2'b01: Y1 = D; // Y1 active when S1=0, S0=1
      2'b10: Y2 = D; // Y2 active when S1=1, S0=0
      2'b11: Y3 = D; // Y3 active when S1=1, S0=1
      default: ; // No action
    endcase
  end
endmodule
```

SIMULATION RESULTS.

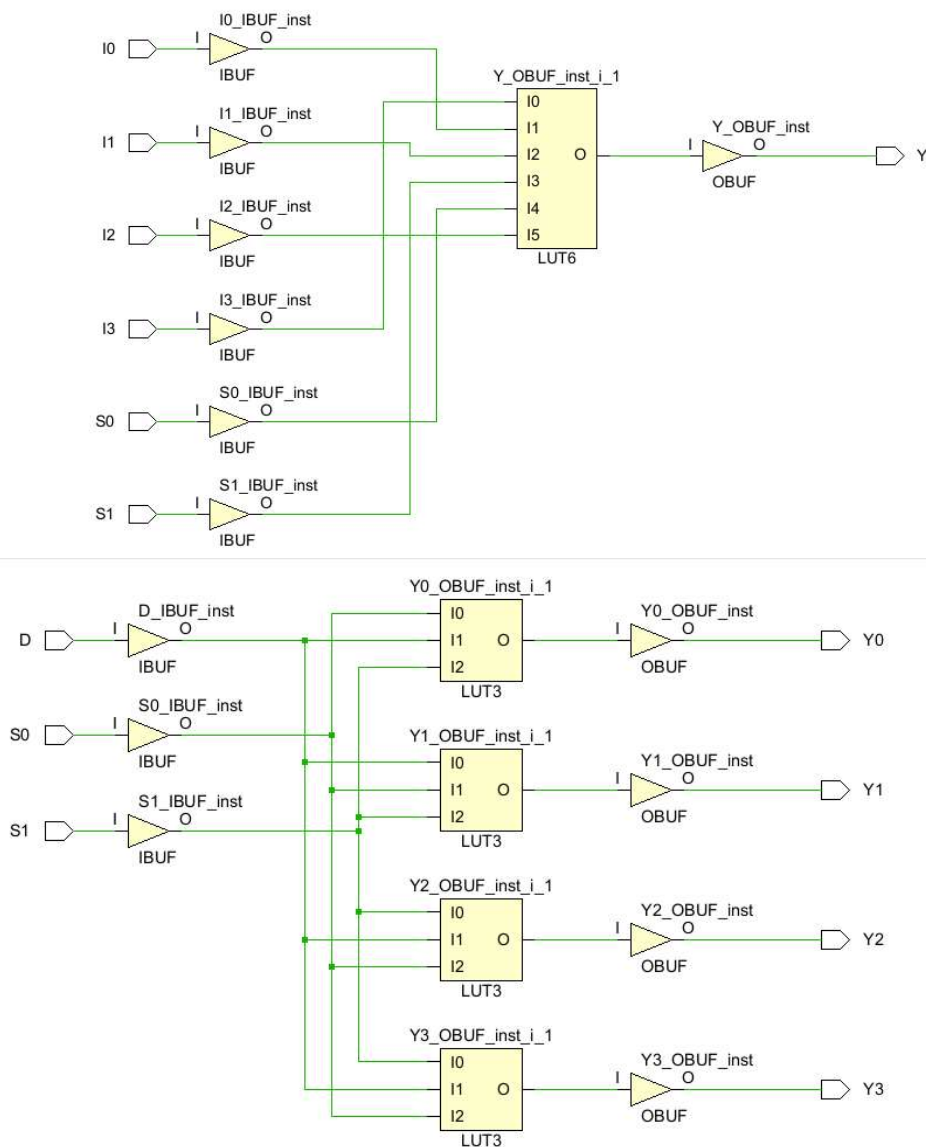
MUX:



DEMUX:



SYNTHESIS MODEL.



CONCLUSION.

The 4:1 MUX and 1:4 DEMUX were successfully designed and simulated in **Vivado** on the **Spartan-7 FPGA**. The MUX correctly selected one input based on the select lines, and the DEMUX sent the input to the correct output. The design worked as expected, proving the logic and functionality were correct.

POST LAB QUESTION.

1. Derive the Boolean equation for output of 4:1 MUX & 1:4 D-MUX.

• 4:1 MUX

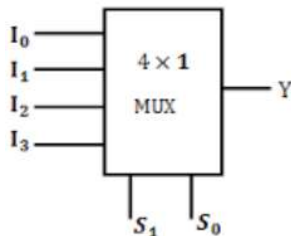
In the Given Block Diagram I₀, I₁, I₂, and I₃ are the 4 inputs and Y is the Single output which is based on Select lines S₀ and S₁.

When S₁S₀=00, the input I₀ is selected.

When S₁S₀=01, the input I₁ is selected.

When S₁S₀=10, the input I₂ is selected.

When S₁S₀=11, the input I₃ is selected.



Truth Table:

Truth Table

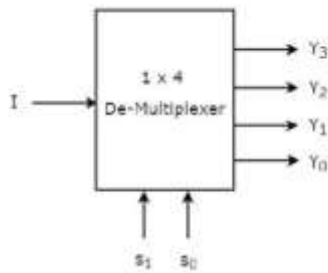
S ₀	S ₁	Y
0	0	I ₀
0	1	I ₁
1	0	I ₂
1	1	I ₃

So, final equation,

$$Y = S_0'.S_1'.I_0 + S_0'.S_1.I_1 + S_0.S_1'.I_2 + S_0.S_1.I_3$$

• 1:4 DEMUX

The 1x2 Demultiplexer consists of 1 input line (I), 1 select line (S), and 2 output lines (Y₀ and Y₁)



Select Line		Outputs			
S_1	S_0	Y_3	Y_2	Y_1	Y_0
0	0	0	0	0	I
0	1	0	0	I	0
1	0	0	I	0	0
1	1	I	0	0	0

For $S_1S_0 = 00 \rightarrow Y_0$ gets D

For $S_1S_0 = 01 \rightarrow Y_1$ gets D

For $S_1S_0 = 10 \rightarrow Y_2$ gets D

For $S_1S_0 = 11 \rightarrow Y_3$ gets D

$$Y_0 = \bar{S}_1 \bar{S}_0 I$$

$$Y_1 = \bar{S}_1 S_0 I$$

$$Y_2 = S_1 \bar{S}_0 I$$

$$Y_3 = S_1 S_0 I$$

ASSESSMENT RECORD.

Practical Performance [Model/Code, Results] (5 Marks)	Presentation/ Learning Outcome [Observation, Conclusion] (5 Marks)	Total (10 Marks)	Sign